

Assignment8: Azure Disk Storage

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Part A: Theoretical Questions

1. What is Azure Disk Storage?

Azure Disk Storage is a block-level storage service for Azure VMs, providing durable, high-performance storage.

2. Azure Disk Types:

- Standard HDD: Cost-effective for infrequent data access.
- Standard SSD: Reliable for general-purpose workloads.
- Premium SSD: High performance for I/O intensive apps.
- Premium SSD v2: Advanced scalability and performance.
- Ultra Disk: Maximum performance and low latency.

3. Difference between OS Disk & Data Disk:

OS Disk hosts the boot operating system, while Data Disk is an extra disk used for storing application data/files.

4. Can multiple VMs share the same disk?

Generally, Azure managed disks are attached to a single VM, but specific features like "Shared Disks" allow sharing for specific scenarios (like SQL Failover clusters).

5. Can disks be resized?

Yes, Azure disks can be resized after creation without deleting the disk.

6. Throughput and IOPS:

IOPS (Input/Output Operations Per Second) defines the number of read/write operations, while Throughput defines the data transfer rate (MB/s).

7. LRS vs ZRS:

LRS (Locally Redundant Storage) replicates data within one datacenter. ZRS (Zone Redundant Storage) replicates data across availability zones.

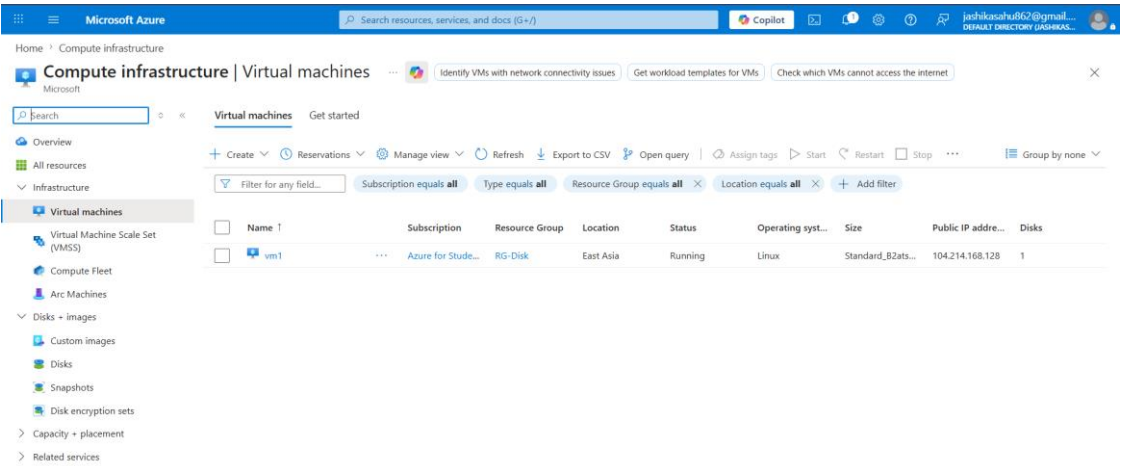
8. Why is disk encryption important?

It protects data at rest, ensuring that even if the storage hardware is compromised, the data remains unreadable.

Part B: PracticalAssignment

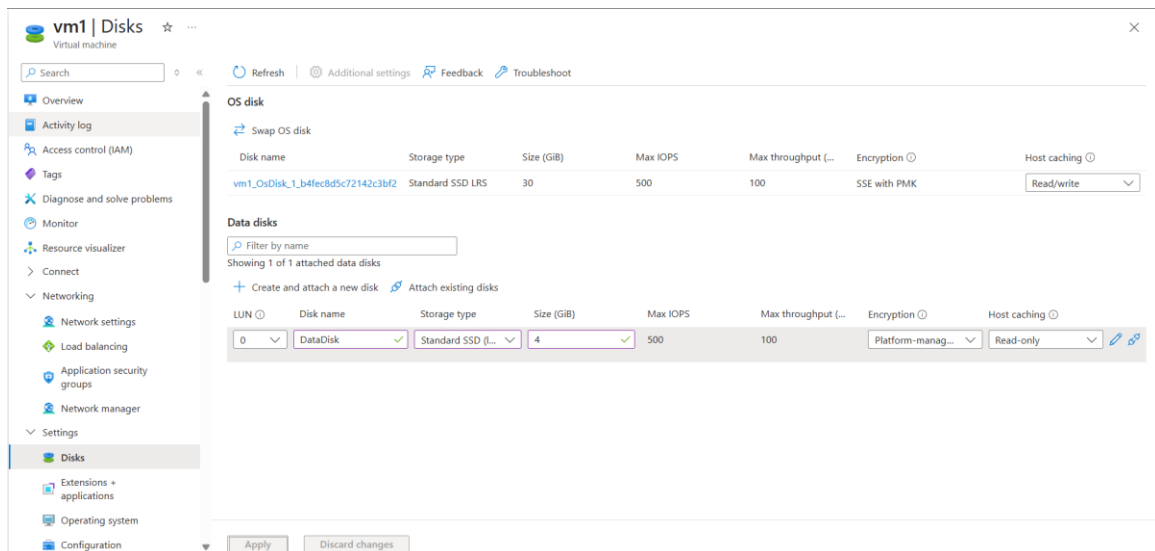
Task1:CreatingVM1

Step:Log in to Azure,createaVirtualMachine,setnameasVM1,and configure credentials.



Task2:Attaching DataDisk

Step:Navigated to VM1Disks, selected 'Create and attach new disk' with 2GiB size.



Task3:Format and Mount

Step:Used SSH to access the VM. Ran `lsblk` to find the disk, then formatted and mounted to `/test`.

```

azureuser@vm1: ~
azureuser@vm1:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0       7:0      0  63.8M  1 loop /snap/core20/2769
loop1       7:1      0  91.7M  1 loop /snap/lxd/38800
loop2       7:2      0  49.3M  1 loop /snap/snapd/26865
sda         8:0      0   30G  0 disk 
├─sda1      8:1      0  29.9G  0 part /
├─sda14     8:14     0    4M  0 part 
└─sda15     8:15     0  106M  0 part /boot/efi
sdb         8:16     0    4G  0 disk 
sr0        11:0     1   628K  0 rom  
azureuser@vm1:~$ sudo mkfs -t ext4 /dev/sdc
mke2fs 1.46.5 (30-Dec-2021)
The file /dev/sdc does not exist and no size was specified.
azureuser@vm1:~$ sudo fdisk /dev/sdc

Welcome to fdisk (util-linux 2.37.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

fdisk: cannot open /dev/sdc: No such file or directory
azureuser@vm1:~$ sudo fdisk /dev/sdb

Welcome to fdisk (util-linux 2.37.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS disklabel with disk identifier 0x7ec0c2c9.

Command (m for help): m

Help:

DOS (MBR)
a  toggle a bootable flag
b  edit nested BSD disklabel
c  toggle the dos compatibility flag

Generic

```

```
azureuser@vm1: ~  
  
Misc  
m print this menu  
u change display/entry units  
x extra functionality (experts only)  
  
Script  
I load disk layout from sfdisk script file  
O dump disk layout to sfdisk script file  
  
Save & Exit  
w write table to disk and exit  
q quit without saving changes  
  
Create a new label  
g create a new empty GPT partition table  
G create a new empty SGI (IRIX) partition table  
o create a new empty DOS partition table  
s create a new empty Sun partition table  
  
Command (m for help): n  
Partition type  
p primary (0 primary, 0 extended, 4 free)  
e extended (container for logical partitions)  
Select (default p):  
  
Using default response p.  
Partition number (1-4, default 1):  
First sector (2048-8388607, default 2048):  
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-8388607, default 8388607):  
  
Created a new partition 1 of type 'Linux' and of size 4 GiB.  
  
Command (m for help):  
  
Command (m for help): n  
  
All space for primary partitions is in use.
```

```
azureuser@vm1:~$ sudo mkfs.ext4 /dev/sdb1  
mke2fs 1.46.5 (30-Dec-2021)  
Discarding device blocks: done  
Creating filesystem with 1048320 4k blocks and 262144 inodes  
Filesystem UUID: 864561ce-0b14-45af-863b-075071478e32  
Superblock backups stored on blocks:  
32768, 98304, 163840, 229376, 294912, 819200, 884736  
  
Allocating group tables: done  
Writing inode tables: done  
Creating journal (16384 blocks): done  
Writing superblocks and filesystem accounting information: done  
  
azureuser@vm1:~$ sudo mkdir /test  
azureuser@vm1:~$ sudo mount /dev/sdb1 /test  
azureuser@vm1:~$ df -h /test  
Filesystem      Size  Used Avail Use% Mounted on  
/dev/sdb1       3.9G  24K  3.7G   1% /test  
azureuser@vm1:~$ echo '/dev/sdb1 /test ext4 defaults 0 0' | sudo tee -a /etc/fstab  
/dev/sdb1 /test ext4 defaults 0 0  
azureuser@vm1:~$
```

Task4:Resize the Disk

Step:Updated size in Azure Portal to 8GiB, then ran `growpart` and `resize2fs` in Linux to apply changes.

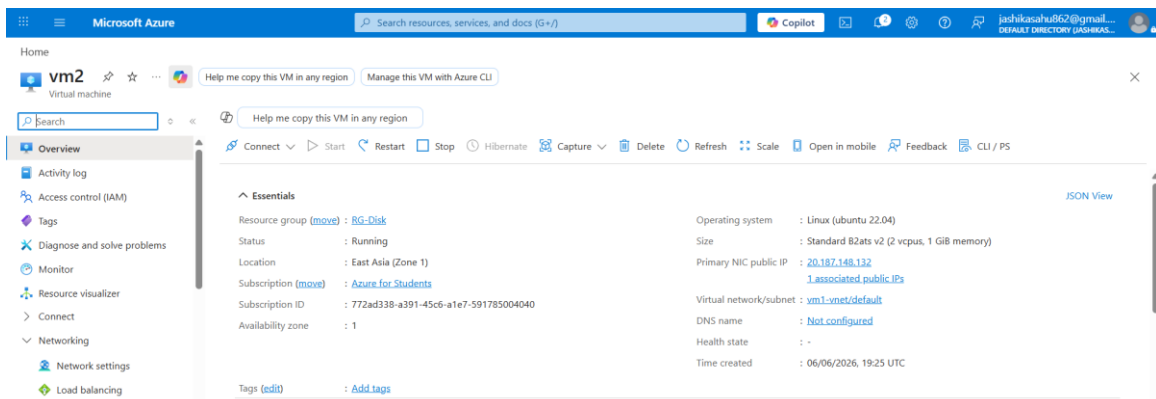
```
azureuser@vm1:~$ sudo apt-get install cloud-guest-utils
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
cloud-guest-utils is already the newest version (0.32-22-g45fe84a5-0ubuntu1).
cloud-guest-utils set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
azureuser@vm1:~$ sudo growpart /dev/sdb 1
CHANGED: partition=1 start=2048 old: size=8386560 end=8388608 new: size=16775135 end=16777183
azureuser@vm1:~$ sudo resize2fs /dev/sdb1
resize2fs 1.46.5 (30-Dec-2021)
Filesystem at /dev/sdb1 is mounted on /test; on-line resizing required
old_desc_blocks = 1, new_desc_blocks = 1
The filesystem on /dev/sdb1 is now 2096891 (4k) blocks long.

azureuser@vm1:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0       7:0      0  63.8M  1 loop /snap/core20/2769
loop1       7:1      0  91.7M  1 loop /snap/lxd/38800
loop2       7:2      0  49.3M  1 loop /snap/snapd/26865
sda         8:0      0   30G   0 disk 
├─sda1      8:1      0  29.9G   0 part /
├─sda14     8:14     0    4M   0 part 
└─sda15     8:15     0  106M   0 part /boot/efi
sdb         8:16     0    8G   0 disk 
└─sdb1      8:17     0    8G   0 part /test
sr0         11:0     1   628K   0 rom 

azureuser@vm1:~$ df -h /test
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdb1       7.8G  24K  7.5G   1% /test
azureuser@vm1:~$
```

Task5:Detach and Attach to VM2

Step:Detached from VM1 and re-attached to VM2 via SAzure Portal.



The screenshot displays the Microsoft Azure portal interface. At the top, the 'Home' page is visible with a search bar and navigation icons. The main content area shows the details for a virtual machine named 'vm2'. The left sidebar contains a navigation menu with options like 'Overview', 'Activity log', 'Access control (IAM)', 'Tags', 'Diagnose and solve problems', 'Monitor', 'Resource visualizer', 'Connect', 'Networking', 'Network settings', and 'Load balancing'. The 'Overview' tab is selected, showing a summary of the VM's configuration. The 'Essentials' section includes details such as the resource group (RG-Disk), status (Running), location (East Asia (Zone 1)), subscription (Azure for Students), subscription ID (772ad338-a391-45c6-a1e7-591785004040), and availability zone (1). The 'Operating system' is Linux (ubuntu 22.04), and the 'Size' is Standard B2ats v2 (2 vcpus, 1 GiB memory). The 'Primary NIC public IP' is 20.187.148.132, and the 'Virtual network/subnet' is vm1-vnet/default. The 'DNS name' is Not configured, and the 'Health state' is -. The 'Time created' is 06/06/2026, 19:25 UTC. The right sidebar shows a 'JSON View' button.

```
azureuser@vm2:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0       7:0      0   63.8M 1 loop /snap/core20/2769
loop1       7:1      0   91.7M 1 loop /snap/lxd/38800
loop2       7:2      0   49.3M 1 loop /snap/snapd/26865
sda          8:0      0    30G  0 disk
├─sda1       8:1      0   29.9G  0 part /
├─sda14      8:14     0     4M  0 part
└─sda15      8:15     0   106M  0 part /boot/efi
sdb          8:16     0     8G  0 disk
└─sdb1       8:17     0     8G  0 part
sr0         11:0     1    630K  0 rom
azureuser@vm2:~$ sudo mkdir /test
azureuser@vm2:~$ sudo mount /dev/sdb1 /test
azureuser@vm2:~$ cd /test
azureuser@vm2:/test$ ls
lost+found
azureuser@vm2:/test$
```